Speech Signal Processing

LAB III
Hearing and Speech Perception

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10-min knowledge sharing/discussion

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Make groups

【腾讯文档】Groups - 语音信号处理（2026春）
https://docs.qq.com/sheet/DY1dTSEtkUXVWYkdF?tab=BB08J2

Purpose of this lab...

1. Continue to learn MATLAB functionality for speech processing (adjust SNR level, spectrogram, etc.)
2. Learn to synthesis vowels with pure tones
3. Test loudness/equal loudness perception
4. Learn to test speech quality with PESQ index

SNR

- Signal-to-Noise Ratio (SNR)

$$SNR = 10\log_{10}(\frac{E_{signal}}{E_{noise}})$$

- In MATLAB:

```
SNR=20*log10(norm(sig)/norm(noise))
```

- Add white gaussian noise

```
y = awgn(x, snr, signalpower)
```

- Equalization

```
y_equalized = y/norm(y)*norm(x)
```

```
t = 0:pi/8:6*pi;
x = sin(t);
y = awgn(x, 5, 'measured');
figure; plot(t,[x; y])
y_e=y/norm(y)*norm(x);
figure; plot(t,[x; y; y_e])
```

Spectrogram

spectrogram(x, window, noverlap, nfft, freqloc)

[s, f, t] = spectrogram(x, window, noverlap, f, fs, freqloc)

```
[s, fs] = audioread("mhint_01_01.wav");
spectrogram(s, 10e-3*fs, [], [], fs, 'yaxis');
% 10ms for each window and 50% overlap
```

x: input signal
window: specified window, [] eight segments with hamming window
noverlap: number of overlapped samples, [] 50% overlap
nfft: number of DFT points
f: frequencies
fs: sample rate
freqloc: frequency display axis, ‘yaxis’

Problem 1*

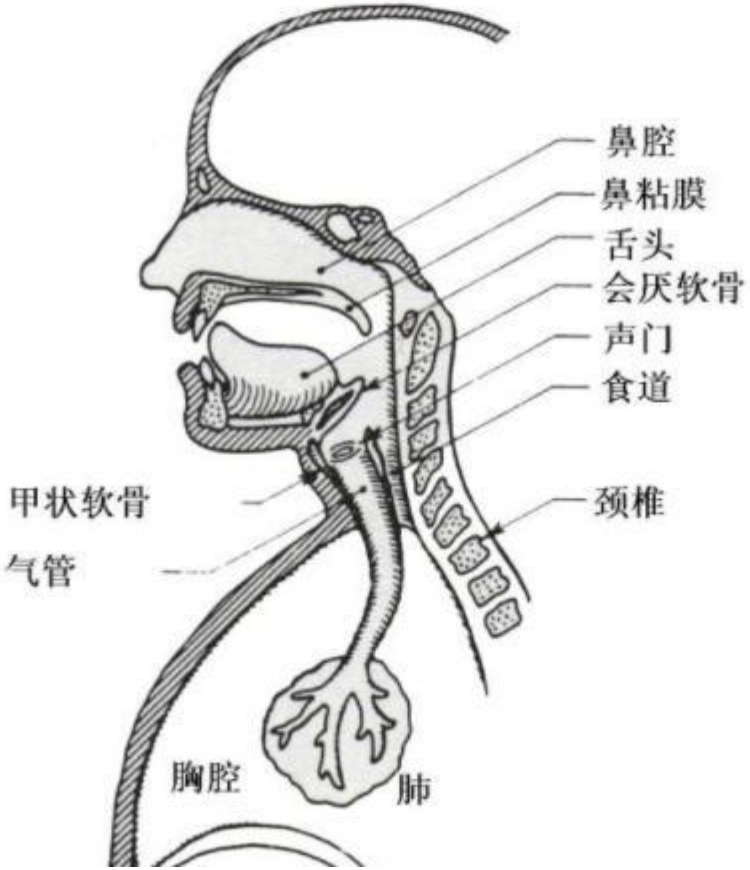
Synthesize vowels using pure tones:

1. Sample the spectral envelopes of vowels /a/ /i/ /u/ in lab 2.
2. Use pure-tones to synthesize vowels /a/ /i/ /u/ (with F0 150 Hz and 250 Hz for male and female voice, respectively).
3. Plot the waveforms/spectra/spectrograms of the 3 synthesized vowels.

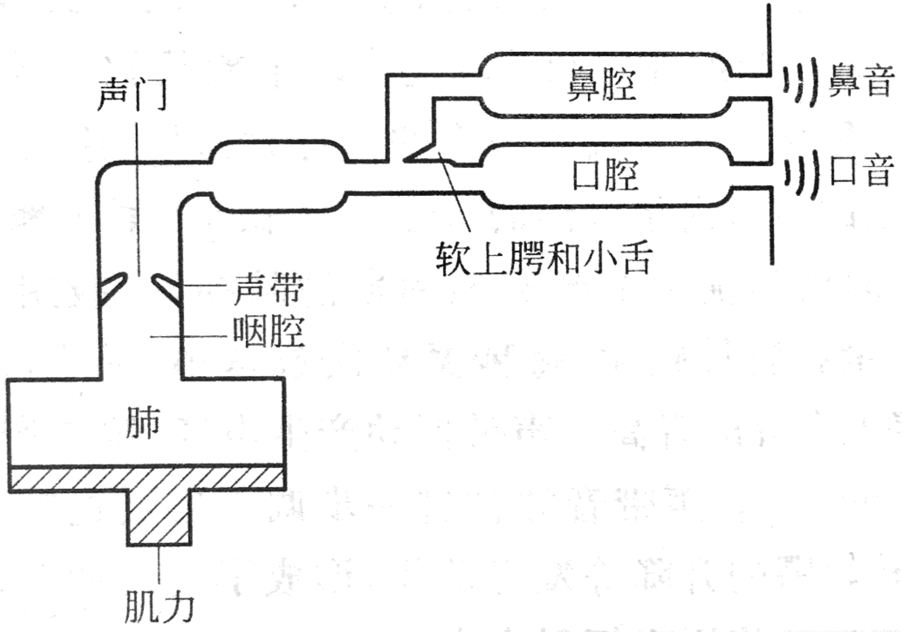
   spectrogram(...)
4. Attach the wav files of synthesized vowels with lab report.

*： bonus

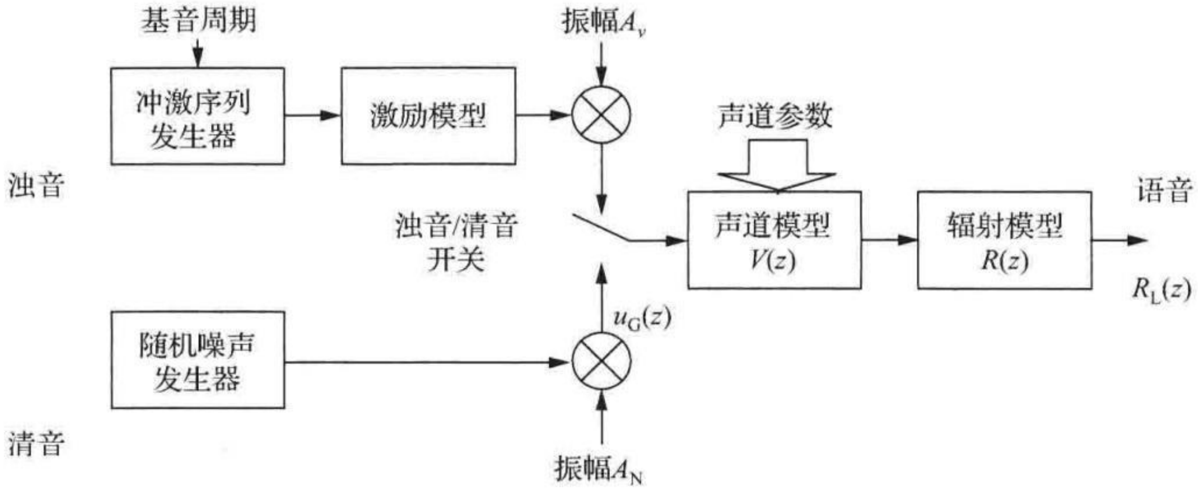
Problem 1*



鼻腔
鼻粘膜
舌头
会厌软骨
声门
食道
甲状软骨
气管
颈椎
胸腔
肺



声门
鼻腔
鼻音
口腔
口音
软上腭和小舌
声带
咽腔
肺
肌力



基音周期
冲激序列发生器
激励模型
振幅A_v
浊音
浊音/清音开关
声道参数
声道模型 $V(z)$
辐射模型 $R(z)$
语音
$R_L(z)$
$u_G(z)$
随机噪声发生器
清音
振幅A_N

Problem 2

http://www.phys.unsw.edu.au/jw/hearing.html

1. Hearing test on-line: sensitivity, and equal loudness contour
2. Screen-copy the equal loudness contour you obtained.

Problem 3

1. For mhint_01_01.wav, generate noisy speech by white noise at 5 and 0 dB SNR.
2. Equalize the energy of noisy speech to that of clean speech.
3. Plot the waveforms and spectra of clean speech and equalized noisy speech at 5 and 0 dB SNR.
4. Call MATLAB command to show spectrograms of clean speech and noisy speech at 5 and 0 dB SNR. spectrogram(...)

Problem 4

Objective speech quality evaluation

1. Add white noise to a clean speech signal (mhint_01_01.wav) at -5, -3, -1, 1, 3, 5 dB SNR level.
2. Equalize the energy of noisy speech to that of clean speech.
3. Run PESQ code in MATLAB and plot the PESQ value as a function of SNR level.

scores = pesq(ref_wav, deg_wav)